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A location-routing-loading problem for bill delivery services

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Abstract

A telecommunication service company in Hong Kong considers printing and delivering its monthly bills to well-structured and densely populated housing estates by the in-house delivery team. There is a large customer base that prefers receiving paper bills for personal or company accounting purposes. The use of in-house delivery team would be more cost-effective than the existing practice of employing the Post Office service. With the anticipated increase of housing estates in the newly developed areas, management is considering to relocate the existing bill delivery office and set up some delivery depot sites at the existing company buildings. Metaheuristic approach based on threshold accepting and simulated annealing is developed to assist in making facility location, vehicle routing and loading decisions. The computerized algorithm has been tested by the operation team and is well perceived by the management. It shows significant improvement over the existing manual approach and will be a tool useful for planning future office locations in Hong Kong. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Location; Routing; Threshold accepting; Simulated annealing; Loading

1. Introduction

A local case of integrated facility location and distribution planning problem is described in this paper. Distribution of bills to customers is a major issue to a telecommunication service company that serves a large customer base of business and residential customers. The company also offers mobile services, Internet and interactive multimedia services. Despite issuing electronic bills, there is still over a million paper bills to be delivered to customers everyday. Besides residential telephone bills, the current mail also include promotional material and mobile phone bills. Many customers still prefer receiving paper bills for personal or company accounting purposes. It is simply a habit adopted from the past, which may change as the society becomes more receptive towards the use of technology. The customers referred

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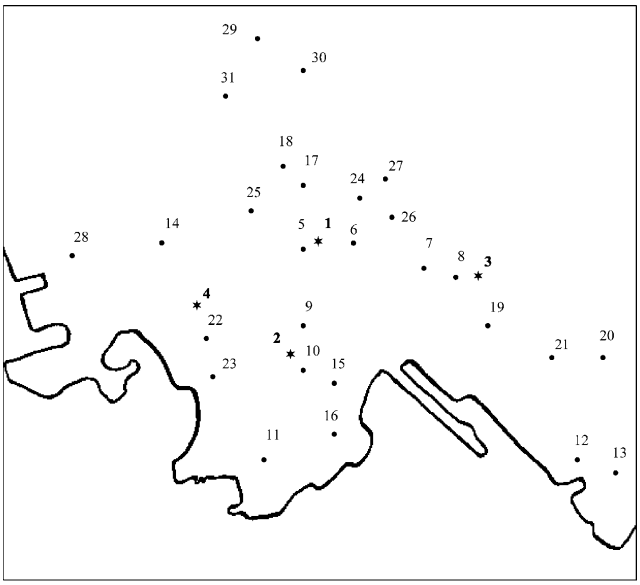


* potential depot location; ♦ housing estate

Fig. 1. The spatial distribution of locations of potential depots and housing estates.

here are mainly residential customers in housing estates who may not have access to computer facilities. The elderly generation mostly prefer paper bills to electronic bills. Besides, as the postage cost is paid by the company and no discount has been offered on switching to electronic bills, the majority of bills are still sent out by post. The faster the bills are delivered to customers, the earlier the company can liquidate in view of cash management. Currently, the monthly postage cost of these bills is over millions of dollars at HK$1.20 per bill. This has imposed a heavy burden in operating expenditures. In future, the management also plan to include non-regular items in the delivery service such as fax directory, yellow pages, calling card, user guides, etc. How to operate the delivery service in the most cost-effective way is becoming increasingly important. As many customers are located in well-structured and densely populated housing estates (see Fig. 1 for those in the Kowloon peninsula), a preliminary study has found that using in-house delivery service can provide cost savings. Besides, it can offer end-to-end service to the customers and achieve higher service standard than through the Post Office.

At present, there is one facility opened in the Hong Kong Island with over 60 staff to handle printing, sorting, packing and delivery of bills. The daily working hours is 8 h. Apart from maintaining its own fleet, vehicles (each manned by a messenger) are also rented to deliver bills to housing estates with a maximum capacity of 5000 bills per vehicle. In Hong Kong, vehicle rental is more cost-effective than purchasing and maintaining the fleet. For the sake of cost estimation in this study, it is assumed that all vehicles used will be rented. The current monthly delivery volume is over 600,000 bills covering 200 public housing estates in Hong Kong. With the anticipated increase of customer size at the new housing estates, the bill volume will be expected to increase accordingly. The existing facility will not have sufficient capacity to cope with future demand. For long-term capacity planning, the management is considering to relocate the existing office and set up some delivery depot sites at the existing 15 potential locations in Hong Kong. A potential depot has a daily capacity of handling 25,000 bills.

At the operational level, the allocation decision of housing estates to the established facilities needs to be made based on the customer demand volume. Good routing and scheduling decisions, subject to the vehicle capacity and working hour constraints, will allow more customers to be served in shorter time. There is currently no time window constraint as customers need not be present during the delivery. If multiple routes could be assigned to a vehicle without exceeding the staff working hours, further cost reduction could be achieved.

Considering the above interdependent decisions, the basic questions to address are: given a set of demand nodes (target housing estates) and a set of potential facilities (delivery depot sites), where should the facilities be established? How should routes be formed from a facility and what is the routing sequence to the demand nodes such that all demand nodes are served? What is the smallest number of vehicles (vehicle fleet size) to be used for a given set of established routes? The primary objective of the above location, routing and loading problems is to minimize the sum of facility costs, staff costs, vehicle rental and operating costs. (The breakdown of the various cost components is given in Table 4.) Naturally, there are other criteria considered by the management level and operational level, such as load balancing of messengers (number of bills carried; actual working hours) and depots (number of bills printed and delivered; number of messengers assigned). Considering that this integrated 3-stage decision problem is relatively novel in literature, we decide to tackle the single cost objective first, which is the primary concern of higher level management. Multiple criteria will be considered in future research.

2. Related literature

Literature on the related vehicle routing problems (VRP) has been numerous (Bramel & Simchi-Levi, 1997; Breedam, 2001; Eilon, Watson-Gandy, & Christofides, 1971; Golden & Assad, 1988), as compared with those on the integrated problems. Routing is nevertheless a necessary component for location and distribution problems. Exact approaches to VRP include mixed integer programming, branch and bound, dynamic programming and set partitioning algorithms (Laporte, 1988; Min, Jayaraman, & Srivastava, 1998). All of them suffer from the curse of dimensionality. Dynamic programming suffers from the drawback of large number of state spaces and similarly, set partitioning algorithm involves large number of variables, even for small problems. Improvement procedures are often designed to streamline the exact algorithm for the specific case considered.

As the location problem and vehicle routing problem have been shown to be NP-hard, respectively, thus the location-routing problem (LRP) is also NP-hard (Tuzun & Burke, 1999). The classification of LRP problem with regard to its problem perspective or solution method can be found in a recent survey (Min et al., 1998). The general LRP can be formulated as an integer or mixed integer program involving three-index variables defining the node-to-node connection and route assignment (Hansen, Hegedahl, Hjortkjær, & Obel, 1994; Perl & Daskin, 1985; Tuzun & Burke, 1999). Capacity constraints on vehicle (and facility) are considered. Apart from capacitated vehicles and facilities, an additional constraint we consider here is the trip time limit of vehicle, imposed by the daily staff working hours. An earlier LRP survey (Laporte, 1988) also presented some practical applications and details about some exact algorithms for LRP. For some unconstrained versions of LRP, where not all the capacity constraints of vehicle and facility, and trip time limit exists, exact integer programs coupled with constraint relaxation can solve up to 50 demand nodes (Laporte, 1988; Laporte, Nobert, & Pelletier, 1983). It was mentioned that the efficiency of the constraint relaxation methods diminished in the case of highly constrained

problems. The excessive computational time on large problems and the additional effort used to build user-friendly interface renders it impractical for implementation. Heuristic approach has been commonly adopted. These include deterministic heuristics based on location-allocation and savings concepts. Madsen (1983) examined a two-level newspaper delivery problem that considers locations of factory, transfer points and routing decisions to customers. Two procedures of alternate location-allocation-savings and savings-drop procedure are recommended. Other iterative approach includes opening (closing) all depots initially, and then systematically closing (opening) depots one by one based on cost savings (Hansen et al., 1994; Srivastava, 1993). Min (1996) considered a two-level location-allocation problem of terminals to customer clusters and supply sources by use of an exact integer program. Routing is determined by a travelling salesman (TSP) algorithm as the vehicle capacity constraint has been handled in the location-allocation phase. The hierarchical approach thus consists of both exact and heuristic procedures. To simplify the LRP, approximation formula to estimate the distribution cost (vehicle fleet cost and routing cost) has been applied under certain hypotheses for a multi-period dynamic LRP (Laporte & Dejax, 1989). However, these hypotheses of convex, compact user-space, uniform distribution of users (demand nodes) in this space, and Euclidean travel metric do not necessarily hold in our case. Decomposing the LRP into subproblems and then solving the subproblems either optimally or heuristically (Perl & Daskin, 1985; Srivastava, 1993) is also a common approach and is adopted in our study. Recently, metaheuristic approach has been applied to tackle the integrated problem (Tuzun & Burke, 1999). While the decomposition and metaheuristic approach will be the focus of our work, a branch and bound procedure using the heuristic results as input is proposed as an exact algorithm for performance evaluation of heuristics.

3. Heuristic approach

The most popular approach in tackling LRP has been reported to be decomposition, followed by a savings method (Min et al., 1998). Examples of such are location-allocation-first, route-second and route-first, and location-allocation-second. However, the limitation of the sequential decomposition approach does not consider the tradeoffs between location and routing decisions. The heuristic design here attempts to tackle the integrated problem iteratively, by employing metaheuristics and exploiting today's computational power. Fig. 2 depicts the flowchart of the methodology. The entire problem is divided into three phases: facility location, routing and loading. The three phases will be tackled repeatedly for a set of facilities of minimal size until the total costs justify the algorithm termination. In each phase, either problem-specific heuristics or an optimal algorithm is applied.

Initially, the pre-processing stage involves transforming the various time components in this problem into the node-to-node travel time parameters adopted in classical LRP literature. There are two components of time requiring data collection: travelling time between housing estates and stopover time (sum of bill delivery time and layover time in the housing estate). The travelling time is estimated by the driver from his/her experience travelling along the road during the work duty. He/she will consider the non-congested road/highway/tunnel/flyover in order to save time. (In the long run, this can be collected from some geographical information systems.) The stopover time components are also supplied by the operational staff. The bill delivery time depends very much on whether the building has mailbox at the ground floor lobby. If there is no mailbox in the lobby such as in the old estate type, the messenger needs to walk to each floor and each door to deliver the mail. The layover time depends on the size of the estate,

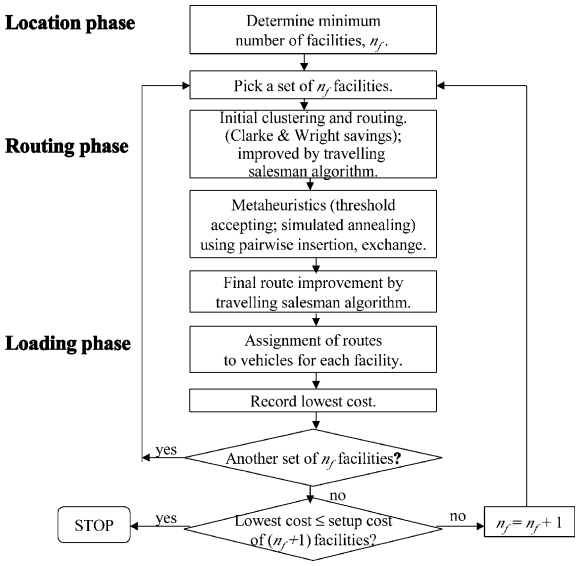


Fig. 2. Flowchart of heuristic design.

which in turn depends on number of blocks in the estates. Let s_i denote the stopover time at node i, and t_{ij} the travelling time between node i and node j in the set of facilities and demand. Then the transformed travelling time, t'_{ij}, between node i and node j is given by:

$$t'_{ij} = t_{ij} + \tfrac{1}{2}(s_i + s_j) \qquad \forall i \neq j \tag{1}$$

In this way, the problem is converted to the classical LRP problem with no stopover time considered. As each vehicle is manned by one messenger (or staff), these two terms will be used interchangeably in this paper. The same applies to the terms 'facility' and 'depot'.

3.1. Number of facilities

The first procedure in the location phase is to determine a lower bound on the number of facilities (n_f) to establish. This is derived from the customer demand (q_i) and the maximum capacity of a facility (Q_f):

$$n_f = \left\lceil \frac{\sum_{i \in H} q_i}{Q_f} \right\rceil \tag{2}$$

where H is the set of demand nodes and $\lceil x \rceil$ is the smallest integer larger than or equal to x. For the current bill delivery problem, the target demand nodes are housing estates in well-structured and densely populated areas so that it is more cost-effective for in-house delivery. Remaining bills in remote areas will be sent via Post Office as usual. Hence, the calculation of n_f can ignore the spatial distribution of demand nodes and facility location. In the selection of the n_f locations, those with more nearby customers will be evaluated first. For every set of n_f facilities selected, the routing and loading phases will be performed. Considering that the number of facilities to open is comparatively smaller than the

number of demand nodes, the complete enumeration of all combinations of n_f locations is fast with today's computing power and the efficient design of the subsequent algorithms. Besides, the printed solution for every set of n_f locations will enable management to examine performance (as demand changes across periods) in order to make long-term facility decisions.

3.2. Initial clustering and routing

The routing phase starts with the initial assignment of customer demand nodes to facilities. The nearest facility for each demand node is first identified. Then a set of n_f facilities will be chosen in descending order of the number of nearest demand nodes. Initial clustering is formed by assigning demand nodes to the nearest open facility that has sufficient handling capacity. If any demand node is left unassigned, it will be assigned to the facility with the least workload although the initial facility capacity will be exceeded. Later reshuffling of assignment by local search will tend to reduce the gap in workload among the facilities. Clarke and Wright (1964) savings algorithm with capacity constraints (of depot and vehicle) and trip time constraint is applied to establish the initial routes for each facility considered. The main strength of Clarke and Wright's algorithm is its simplicity and robustness. It is simple enough to be easily modified to include other constraints and possible time window constraints in future, and robust enough to give a reasonably good solution that can be implemented in practice (Chopra & Meindl, 2001). The routes formed are further improved by a TSP algorithm to determine the best sequence of nodes in each route. Clarke and Wright's algorithm runs in polynomial time. Since the demand size (of a housing estate) is comparable with the vehicle capacity (maximum number of bills to be carried in a route), the number of nodes in a route is not very large. Hence, the TSP algorithm also runs fast even though the problem is NP-hard.

3.3. Improvement by metaheuristics

To improve on the routing cost and vehicle/staff associated costs for the given set of n_f facilities, metaheuristic approach (Osman & Kelly, 1996) of threshold accepting (TA) and simulated annealing (SA) is adopted. TA (Dueck & Scheuer, 1990; Dueck 1993) was introduced as a deterministic version of SA (Kirkpatrick, Gelatt, & Vecchi, 1983). The underlying design is to apply a single or multiple local search methods to obtain a near optimum solution. Using single method simplifies the choice of the design parameters. Building hybrid metaheuristics to combine the best features of both metaheuristics on routing problems has been suggested in past literature (Breedam, 2001). Moreover, using multiple methods avoids being trapped at a local optimum due to the acceptance criterion specific for a single metaheuristic method. When applying multiple methods, the current solution at the termination of one method will be adopted as the starting solution for the subsequent method, and the best found solution will be retained. The acceptance criterion for new solutions in TA (Section 3.5) and SA (Section 3.6) are simple to code without using much memory structure. However, the computational efficiency of the procedure to generate a new solution depends on the complexity of the neighbourhood structure defined.

3.4. Neighbourhood structure

Like other metaheuristics, both TA and SA require the design of a neighbourhood structure and the acceptance criterion of a new solution. The local search starts off with some initial solution and will

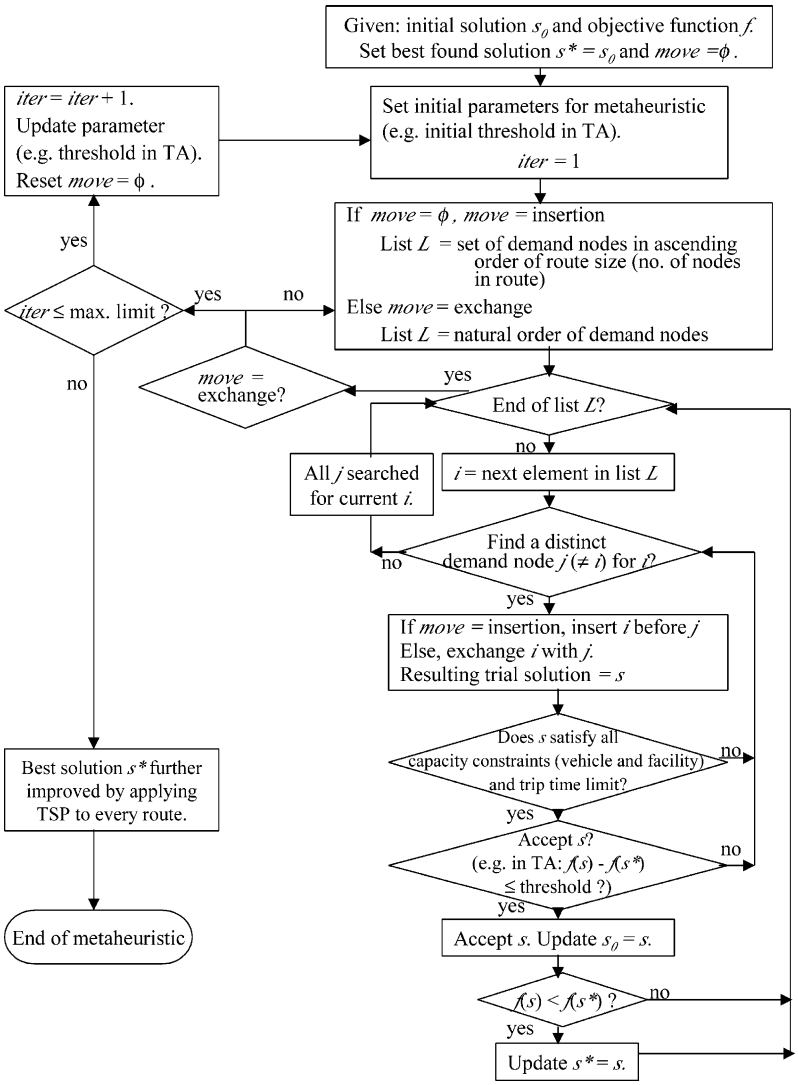


Fig. 3. Metaheuristic approach in the routing phase.

continuously move onto promising neighbourhood solutions. The procedure described here will be summarized in the flowchart of Fig. 3.

Past literature on LRP (Hansen et al., 1994; Tuzun & Burke, 1999) have adopted two types of move to search for neighbouring solutions: insertion and exchange. In each type of metaheuristic here, a set of insertion moves on all pairs of demand nodes is first performed. All demand nodes are first sorted in ascending order of number of demand nodes in their route. The insertion move will examine the change when a demand node in the list is inserted before another demand node, given that the capacity constraints (of route and facility) and the trip time limit are not violated. (The sorted list enables nodes in smaller routes to be considered first, with the aim of eliminating the route by gradually transferring nodes to other routes.) If the move results in a new solution accepted by the metaheuristic

(according to its acceptance criterion), it will be performed. Otherwise, insertion of another pair of demand nodes will be considered. In a similar manner, the exchange move exchanges positions of every pair of demand nodes, provided that the capacity constraints and trip time limits are respected. Notice that the exchange move does not alter the number of nodes in the routes and hence no initial sorting of demand nodes is necessary.

3.5. Threshold accepting

The design of TA provides tolerance for the local search to accept worse solutions initially. Such tolerance will gradually decrease as the search proceeds. The classical TA accepts a new solution if the change in the objective value from the current solution falls within a predetermined threshold. Initially, the threshold is large to allow accepting inferior solutions in order to avoid being trapped in local optima. Thereafter, the threshold will reduce gradually until it becomes a descent algorithm towards the end of the run. The acceptance criterion for new solutions is based on the choice of a (decreasing) sequence of threshold values. This is determined by initial fine-tuning the parameters for the specific problem.

The TA algorithm specific for this problem is outlined in the following:

Select an initial solution s_0 (Section 3.2) and set best $s^* = s_0$
Repeat
  Select a neighbouring solution s (Section 3.4)
  Accept s if
    $f(s) - f(s^*) <$ threshold T. Update $s_0 = s$
    If $f(s) < f(s^*)$, update $s^* = s$
  Update T after a predetermined number of moves (Fig. 3)
Until a predetermined stopping condition is satisfied (Fig. 3)

We adopted a variant of the classical TA by making acceptance decisions based on the change in partial objective value (i.e. $f(s) - f(s^*)$) from the current best solution s^*, rather than from the current solution s_0 as in classical TA. The partial objective value f here refers to the non-facility related costs, which is the sum of routing costs, vehicle rental costs and messenger salary. The facility costs, which are significantly larger than other costs, are left out as initial testing indicates that threshold values based on the smaller cost components result in better solution quality. Besides, the facility costs are invariant for every set of n_f facilities examined. Preliminary testing indicates that this variant performs better than the classical version. The threshold sequence is designed as follows:

$$\text{Threshold } T = p \times (\text{current best partial objective value}) \tag{3}$$

where $p \geq 0$. Initially, the algorithm would allow accepting worse solutions within a certain fraction $p(= p_0 > 0)$ off the current best partial objective value. The threshold value is changed after applying insertion and exchange on all pairs of demand nodes (Fig. 3). As the search proceeds, p will decrease linearly until it reaches the minimum value of 0, i.e. no worse solutions will be accepted. This occurs after some *p_num* steps (= 'max. limit' in Fig. 3) in which the algorithm becomes a local search descent procedure. To summarize, the updating procedure for the

threshold T involves first updating p:

$$p \rightarrow p - \frac{p_0}{p_num} \tag{4}$$

followed by updating T in Eq. (3).

3.6. Simulated annealing

The conceptual design of SA has been known to originate from the physical annealing process of solids (Kirkpatrick et al., 1983). The solid (crystal) is first heated to a high temperature at which all molecules randomly arrange themselves into a liquid phase. Then it is allowed to cool by carefully controlling the temperature change in order to observe the change in its internal structure. This process of careful annealing at which the temperature decreases slowly at a desired rate sets the underlying design of the acceptance criterion of new solutions in SA. Initially, a set of temperature parameters $\{T_1,\ldots,T_K\}$ is selected after fine-tuning for the problem of interest, where length K is also a parameter to be determined. SA proceeds in a way similar to TA. The difference lies in the acceptance criterion for worse solutions, which is probabilistic. When evaluating every new solution, a uniform random number r in [0,1) is generated. Let $\Delta(=f(s)-f(s_0))$ denote the change in partial objective value (non-facility related costs) of the new solution s from the current solution s_0 and T_i, the current temperature value. The new solution will be accepted if

$$r \leq \mathrm{e}^{-\Delta/T_i} \tag{5}$$

Otherwise, it will be rejected and a new neighbouring solution s will be selected. In other words, a probability of $\mathrm{e}^{-\Delta/T_i}$ is allowed for accepting the new solution. The initial temperature value will be set high (driving acceptance probability high) and as the algorithm proceeds, it will gradually decrease. At smaller temperature values, the chance of accepting inferior solutions decreases, which is analogous with the design of TA. Here, we select a cooling rate and boundary temperatures such that initially all inferior solutions will be accepted and towards the end, the probability of accepting a solution with one unit worse off ($\Delta = 1$) remains slightly over 0.1. The temperature sequence is adopted from Lundy and Mees (1986) in which successive temperatures are given by

$$T_{i+1} = \frac{T_i}{1+\beta T_i}, \qquad i = 1,\ldots,K-1 \tag{6}$$

β is the rate parameter expressed in terms of the initial input parameters T_1, T_K and K:

$$\beta = \frac{T_1 - T_K}{(K-1)T_1T_K} \tag{7}$$

Similar to the change in threshold value, the temperature value is changed (by Eq. (6))) when insertion and exchange have been attempted on every pair of demand nodes.

Each of the earlier method will be applied on its own or one after the other, resulting in four different algorithms. TA and SA refer to the single algorithm threshold accepting and simulated annealing, respectively. (TA, SA) represents applying TA first, followed by SA. Similarly, (SA, TA) is the algorithm in reverse order of (TA, SA). The total number of iterations is set to be the same in each algorithm. As no local search method can be proved to be superior over another for all instances of a

problem, it is conjectured that the use of two different methods will incorporate the best features of each and that one will compensate for the deficiency of another.

3.7. Final improvement by travelling salesman algorithm

At the end of the routing phase for the n_f candidate facilities, each route is improved by the travelling salesman algorithm to reduce routing costs while the number of bills in the route remains unchanged. This procedure has been commonly adopted in past literature (Hansen et al., 1994; Min, 1996; Srivastava, 1993). Here, it is not applied during the insertion or exchange moves in order to preserve the current solution structure from being destroyed by greedy improvement in every move of the local search. Even though the TSP is NP-hard, the current LRP has specific characteristics of significant stopover time (due to the large number of bills to be delivered in a housing estate) as compared with travelling time. This results in routes with usually not more than eight demand nodes and hence TSP could be solved efficiently to optimality.

3.8. Routes assignment—a loading problem

When the routing phase is completed for every set of n_f facilities, further vehicle cost reduction could be achieved by assigning multiple routes to a vehicle as long as the total trip time does not exceed the trip time limit. This is the classical vehicle loading problem (Bramel & Simchi-Levi, 1997; Eilon et al., 1971) for items (bills) and vehicles of identical capacity (trip time limit). The objective is to minimize the number of vehicles used. As the number of routes is usually not more than ten under the current environment, the loading problem could be solved optimally by enumerating all possible assignment for each open facility. This is computationally efficient by exploiting today's computational power. On the other hand if the problem size were large, the well-known heuristic, best-fit-decreasing (Bramel & Simchi-Levi, 1997), which is of polynomial time could be applied.

At the end of the loading phase, a complete solution for the current set of n_f facilities is obtained. The procedure repeats again for another set of n_f facilities. If all sets of n_f facilities are explored, and the lowest total cost recorded is not greater than the setup costs of $(n_f + 1)$ facilities, the algorithm will terminate. Otherwise, the process repeats for sets of $(n_f + 1)$ facilities. In real life data, the facility setup cost is usually much higher than other cost components. Hence, the process of adjusting the size of n_f and repeating the three phases will not occur many times. The best solution under each candidate set of facilities and the overall best are displayed as useful information to assist management in making short- or long-term location, routing and loading decisions.

4. Performance evaluation and comparison

The algorithms will be tested on demand nodes (housing estates) and potential depot locations in the Kowloon area as full data collection covering all the target housing sites in Hong Kong is not yet available. There are four potential depots and 27 demand nodes. The time-related data and bill volume for individual sites are given in Table 1. The travelling times between different pairs of nodes are given in Tables 2 and 3. Table 4 records the cost information. Instances taken from a partial data set or simulated from the existing data are created for evaluation of the heuristic. The heuristic solution quality

Table 1
The potential depots and demand nodes (housing estates) for Kowloon

Node index	District	Housing estate/Depot	No. of blocks	No. of bills	Bill delivery time (min) $T1$	Layover time (min) in estate $T2$	Total time (min) $T1 + T2$
$P1$	Kowloon City	Kowloon City Depot					
$P2$	Mongkok	Mongkok Depot					
$P3$	Ngau Tau Kok	Ngau Tau Kok Depot					
$P4$	Shamshuipo	Shamshuipo Depot					
$P5$	Kowloon City	Grandview Garden	4	260	17	6	23
$P6$	Kowloon City	Jubilant Place	7	450	30	11	41
$P7$	Kowloon Bay	Richland Gardens	22	2960	197	33	230
$P8$	Ngau Tau Kok	Amoy Garden	19	2449	163	29	192
$P9$	Ho Man Tin	Chun Man Court	12	868	58	18	76
$P10$	Mongkok	Shun King Building	2	208	14	3	17
$P11$	Yaumatei	Prosperous Garden	4	448	30	6	36
$P12$	Yau Tong	Hong Pak Court	7	1205	80	11	91
$P13$	Yau Tong	Laguna City	38	4037	269	57	326
$P14$	Cheung Sha Wan	Hang Chun Court	2	370	25	3	28
$P15$	Hunghom	Laguna Verde	15	1287	86	23	109
$P16$	Hunghom	Whampoa Estate	25	1368	91	38	129
$P17$	Wong Tai Sin	Chuk Yuen Estate	8	1897	126	12	138
$P18$	Wong Tai Sin	Tsui Chuk Garden	14	1749	117	21	138
$P19$	Kwun Tong	Hiu Lai Court	8	2437	162	12	174
$P20$	Lam Tin	Sceneway Garden	17	2067	138	26	164
$P21$	Lam Tin	Tak Tin Estate	7	1709	114	11	125
$P22$	Shamshuipo	Crown Garden	7	368	25	11	36
$P23$	Shamshuipo	Manor Centre	7	144	10	11	21
$P24$	Choi Hung	King Shan Court	6	792	53	9	62
$P25$	Kowloon Tong	Kowloon Tong Garden	2	62	4	3	7
$P26$	Diamond Hill	Galaxia	5	565	38	8	46
$P27$	Diamond Hill	Lung Poon Court	7	617	41	11	52
$P28$	Lai Chi Kok	Mei Foo Sun Chuen	196	3301	220	157	377
$P29$	Shatin	City One Shatin	52	2135	142	42	184
$P30$	Shatin	Golden Lion Garden	13	1384	92	20	112
$P31$	Shatin	Sui Wo Court	9	1916	128	14	142

will be compared in two criteria (total cost, computational time) with an exact branch and bound algorithm in five smaller problems. As the current problem of four depots and 27 demand nodes requires excessive computational time from the exact method, the manual solution by the Operation Manager will also be provided for comparison with the heuristics.

4.1. Branch and bound

Exact approaches for LRP consist mainly of mathematical programming and branch and bound algorithms (Min et al., 1998). The former often involves incorporating appropriate relaxation procedures or Gomory cuts tailored to the specific LRP. To the authors' knowledge, the existing (mixed) integer programming techniques have not considered solving LRP with capacitated vehicles, capacitated

Table 2
The travelling time between the potential depots and estates $P5,\ldots,P15$ (minutes). '999' is an arbitrary large number imposed to forbid linking of a node to itself

	Potential depots								Housing estates						
	$P1$	$P2$	$P3$	$P4$	$P5$	$P6$	$P7$	$P8$	$P9$	$P10$	$P11$	$P12$	$P13$	$P14$	$P15$
$P1$	999	52	66	57	40	43	72	69	53	51	55	79	73	63	45
$P2$	52	999	55	49	44	46	48	55	42	35	47	69	75	49	52
$P3$	66	55	999	52	57	51	39	36	49	58	65	51	55	59	65
$P4$	57	49	52	999	59	64	59	49	56	42	48	62	66	39	46
$P5$	40	44	57	59	999	35	58	62	49	46	52	70	72	59	42
$P6$	43	46	51	64	35	999	55	59	46	43	49	67	69	56	48
$P7$	72	48	39	59	58	55	999	40	59	55	61	55	58	69	60
$P8$	69	55	36	49	62	59	40	999	64	59	67	51	54	72	67
$P9$	53	42	49	56	49	46	59	64	999	49	58	69	72	47	49
$P10$	51	35	58	42	46	43	55	59	49	999	40	67	64	49	54
$P11$	55	47	65	48	52	49	61	67	58	40	999	71	76	44	51
$P12$	79	69	51	62	70	67	55	51	69	67	71	999	38	62	67
$P13$	73	75	55	66	72	69	58	54	72	64	76	38	999	60	64
$P14$	63	49	59	39	59	56	69	72	47	49	44	62	60	999	54
$P15$	45	52	65	46	42	48	60	67	49	54	51	67	64	54	999
$P16$	49	54	69	54	46	43	62	69	51	57	59	64	67	59	38
$P17$	54	46	52	51	47	44	42	46	57	59	52	54	58	49	57
$P18$	58	51	57	54	51	48	46	49	55	62	57	57	54	47	59
$P19$	71	60	46	59	54	51	41	38	62	69	62	44	44	58	65
$P20$	75	70	49	67	59	56	51	54	68	72	69	42	47	68	69
$P21$	78	76	53	71	63	60	57	55	64	70	66	46	51	72	74
$P22$	59	48	55	38	56	53	62	66	59	51	47	59	57	42	51
$P23$	62	53	58	35	52	49	65	62	53	53	49	63	55	48	54
$P24$	68	59	44	46	55	52	47	49	48	61	62	50	48	58	53
$P25$	51	49	49	52	50	47	49	52	54	53	48	49	59	55	49
$P26$	61	51	43	54	49	46	56	58	63	47	57	54	49	51	59
$P27$	67	56	48	52	53	50	59	62	67	50	64	56	54	53	61
$P28$	67	55	66	40	60	57	69	72	77	69	51	69	64	47	68
$P29$	80	86	90	71	76	73	77	82	69	61	67	74	79	62	78
$P30$	71	80	83	63	70	68	69	74	62	57	62	70	72	60	71
$P31$	76	83	87	67	74	65	72	80	67	59	65	72	78	61	75

facilities and trip time constraint. (Perl and Daskin (1985) presented a mixed integer programming formulation, but recommended a heuristic approach for this difficult problem.) Besides, an additional loading phase (assignment of multiple routes to vehicles) is considered in our problem. Branch and bound algorithms and efficient lower bounds on a single-depot vehicle routing problem using an exact number of vehicles (or routes) have been developed by Christofides, Mingozzi, and Toth (1981). Here, we propose a general branch and bound algorithm which considers M candidate facilities, N demand nodes and at most R routes. It is designed as a depth-first search procedure with the following characteristics:

- maximum number of levels in the search tree: $N + R$

Table 3
The travelling time between the potential depots and estates $P16,\ldots,P31$ (minutes). '999' is an arbitrary large number imposed to forbid linking of a node to itself

	Housing estates															
	$P16$	$P17$	$P18$	$P19$	$P20$	$P21$	$P22$	$P23$	$P24$	$P25$	$P26$	$P27$	$P28$	$P29$	$P30$	$P31$
$P1$	49	54	58	71	75	78	59	62	68	51	61	67	67	80	71	76
$P2$	54	46	51	60	70	76	48	53	59	49	51	56	55	86	80	83
$P3$	69	52	57	46	49	53	55	58	44	49	43	48	66	90	83	87
$P4$	54	51	54	59	67	71	38	35	46	52	54	52	40	71	63	67
$P5$	46	47	51	54	59	63	56	52	55	50	49	53	60	76	70	74
$P6$	43	44	48	51	56	60	53	49	52	47	46	50	57	73	68	65
$P7$	62	42	46	41	51	57	62	65	47	49	56	59	69	77	69	72
$P8$	69	46	49	38	54	55	66	62	49	52	58	62	72	82	74	80
$P9$	51	57	55	62	68	64	59	53	48	54	63	67	77	69	62	67
$P10$	57	59	62	69	72	70	51	53	61	53	47	50	69	61	57	59
$P11$	59	52	57	62	69	66	47	49	62	48	57	64	51	67	62	65
$P12$	64	54	57	44	42	46	59	63	50	49	54	56	69	74	70	72
$P13$	67	58	54	44	47	51	57	55	48	59	49	54	64	79	72	78
$P14$	59	49	47	58	68	72	42	48	58	55	51	53	47	62	60	61
$P15$	38	57	59	65	69	74	51	54	53	49	59	61	68	78	71	75
$P16$	999	47	57	67	70	62	60	51	48	53	50	54	59	72	67	69
$P17$	47	999	37	58	63	67	52	55	56	54	57	59	60	70	62	66
$P18$	57	37	999	56	60	65	50	53	52	50	55	57	58	68	60	63
$P19$	67	58	56	999	45	48	60	57	52	57	55	52	62	78	72	74
$P20$	70	63	60	45	999	39	61	62	53	56	49	46	52	72	62	69
$P21$	62	67	65	48	39	999	66	69	60	62	56	57	58	77	72	75
$P22$	60	52	50	60	61	66	999	37	49	53	57	59	49	62	60	61
$P23$	51	55	53	57	62	69	37	999	52	55	63	62	54	69	62	66
$P24$	48	56	52	52	53	60	49	52	999	54	44	48	59	78	72	74
$P25$	53	54	50	57	56	62	53	55	54	999	55	59	52	61	52	58
$P26$	50	57	55	55	49	56	57	63	44	55	999	38	59	69	61	65
$P27$	54	59	57	52	46	57	59	62	48	59	38	999	62	72	60	66
$P28$	59	60	58	62	52	58	49	54	59	52	59	62	999	62	58	60
$P29$	72	70	68	78	72	77	62	69	78	61	69	72	62	999	45	40
$P30$	67	62	60	72	62	72	60	62	72	52	61	60	58	45	999	52
$P31$	69	66	63	74	69	75	61	66	74	58	65	66	60	40	52	999

- set of nodes: set of candidate facilities $\{f_1,\ldots,f_M\} \cup$ set of demand nodes $\{d_1,\ldots,d_N\} \cup$ set of route dividers $\{r_1,\ldots,r_{R-M}\}$. (The total number of nodes is the same as the total number of levels in the tree.)
- first node of the search tree: always a candidate facility $\in \{f_1,\ldots,f_M\}$
- an arc (i,j) in the search tree: connection between the nodes i and j, which has the physical interpretation depending on the identities of i and j (Table 5)
- branching strategy: select the earliest node in list $L = (f_1,\ldots,f_M, d_1,\ldots,d_N, r_1,\ldots,r_{R-M})$ which has not yet occurred in the partial tree
- feasibility check at each level:
  if a demand node is considered, check the capacity and trip time of the current route (vehicle), and the capacity of the current depot

Table 4
Fixed and variable costs (HK$)

Item	Cost
Fixed cost	
Depot setup cost	$300,000 per depot
Variable cost	
Depot rental cost	$1800 per day
Vehicle rental cost	$1200 per day
Vehicle operating cost	$70 per hour
Messenger salary	$300 per day

if a depot node is considered, check if its capacity can handle the demand of the remaining unvisited demand nodes

- backtracking strategies:
  when no element in L can be assigned to the current level
  when all demand nodes exist in the partial tree (even if the end of the tree has not been reached)
  when backtracking at a level consisting of a route divider, backtrack further up until a demand or facility node is first encountered
- Termination condition: when the first level is reached and the current node is f_M

As explained in Table 5, a facility node in the (partial) tree implies the start of a new facility and a new route, if there exists at least one demand node down the tree before reaching another facility node (or end of tree). Otherwise, this facility is not used. An example of a feasible path in the tree and the corresponding physical solution for an instance of $M = 2$, $N = 10$ and $R = 7$ is shown as follows:

Path: $(f_1, d_1, d_4, d_8, r_5, r_4, d_{10}, d_6, r_3, d_7, d_2, f_2, d_3, r_2, d_9, r_1, d_5)$
Corresponding routes: $(f_1, d_1, d_4, d_8, f_1)$; (f_1, d_{10}, d_6, f_1); (f_1, d_7, d_2, f_1); (f_2, d_3, f_2); (f_2, d_9, f_2); (f_2, d_5, f_2)

The arc (r_5,r_4) in the path implies that a route is empty (no demand node between the two dividers).

To reduce the running time of this branch and bound procedure, the heuristic will be run first and its best solution and properties will be exploited. The branch and bound algorithm will adopt the total cost as the upper bound, and the number of routes as the value for R. (Notice that this is true when the cost of a new route is significantly larger than the vehicle operating cost.) Instead of examining all M candidate facilities in one search, the algorithm will first examine the n_f facilities found in the heuristic solution. Then similar to the heuristic design, it will search for another set of n_f facilities which has the next smaller total cost recorded in the heuristic solution. With its current stage of development, the above branch and bound algorithm can only solve small problems (Section 4.3). However, it provides a framework for the general LRP with potential for improvement, such as those found in Christofides et al. (1981) for VRP.

Table 5
Interpretation of an arc (i,j) in the branch and bound tree search

Identity of node i	Identity of node j	Interpretation of arc (i,j)
Facility	Facility	Facility i is not used, a new facility j is likely to open
	Demand	A new facility i is open and the first route starts with node j
	Route divider	No particular physical interpretation
Demand	Facility	A route is ended with i as the last node, the previous facility has completed its route(s), a new facility j is likely to open
	Demand	Physical arc (i,j) exists in the current route
	Route divider	A route is ended with i as the last node
Route divider	Facility	No particular physical interpretation, but a new facility j is likely to open
	Demand	New route is started with j as the first node
	Route divider	No particular physical interpretation

4.2. Heuristic algorithms and choice of parameters

All the algorithms were coded in Microsoft Visual Fortran Professional 5.0.A and run on a Pentium III 866. After fine-tuning parameters for the current problem instance, the values of the parameters for the metaheuristics are chosen. For the two algorithms utilizing both methods, the initial p_0 value in TA is set to 0.01. This has the interpretation that the initial algorithm allows accepting a worse solution with an increase in cost of no more than 1% (0.01) over the current best solution. p will then decrease in stepsize of p_0/p_num $(= 1 \times 10^{-5})$. After p_num $(= 1000)$ steps, p will be 0 in which the algorithm becomes a descent procedure. For SA, the length of the temperature sequence K, is set to be the same as p_num, i.e. 1000. The initial and final temperatures take on values of $T_1 = 1 \times 10^6$ and $T_K = 0.5$, respectively. These values are chosen such that all worse solutions can be accepted initially. The acceptance probability decreases thereafter and remains slightly over 0.1 for a worse solution with an increase in cost of 1 unit. For the algorithms utilizing a single method, the total number of iterations will be kept the same as the multiple methods. The p value in TA will decrease in stepsize of 0.5×10^{-5} and p_num will be set to 2000. The length of the temperature sequence, K, in SA will be set to 2000. Other parameters remain unchanged.

4.3. Test problems

We investigate the computational performance of the four heuristic algorithms with the branch and bound exact algorithm (Section 4.1) on five test problems before applying them to the large problem. The five problems consist of the current four potential depots and the number of demand nodes ranges from 10–12. The first three problems are subsets of the larger data set (Tables 1–3). The last two problems have data simulated from the existing real data set (e.g. travel times between depot-housing estates and between pairs of housing estates; number of bills in housing estates), while the simulated bill quantities are doubled to represent a future demand increase. The five problems are shown in Table 6. All original system constraints are considered.

Table 7 gives the computational results for the four heuristics and the branch and bound algorithm. There are two problems that the branch and bound does not reach optimality within a time limit of

Table 6
Test problems

Problem	Number of potential depots	Number of demand nodes	Details
1	4	10	Demand nodes $P5$–P 14 (from Tables 1–3)
2	4	10	Demand nodes $P22$–$P31$ (from Tables 1–3)
3	4	12	Demand nodes $P12$–$P23$ (from Tables 1–3)
4	4	10	Simulated data with demand quantity doubled
5	4	12	Simulated data with demand quantity doubled

10,000 CPU s and the best solution found is recorded. Apart from Problem 5, the heuristic solutions coincide with the optimal/best solution in the branch and bound. Using the heuristic solution (selected depots, total cost) as initial conditions does allow the tree search to escape a local minimum, which could not be detected by the heuristics (Problem 5). The tradeoff is the unpredictable length of running time, which increases exponentially when only a few demand nodes is added. The four heuristics do not show significant differences in solution value or computational time in these five problems.

4.4. Comparison with manual approach

As the branch and bound algorithm does not give a feasible solution after long running hours for the larger problem of four depots and 27 demand nodes, another means of comparison with the heuristics is the manual approach and its solution. However, no conclusion can be made about the absolute deviation from optimality, even if one method is significantly better than another.

The company has carried out some initial planning for this problem. According to management planning, the housing estates are clustered into districts and then further clustered into zones. The approach taken corresponds to a 2-layer cluster-first-route-second method. There are two zones for Kowloon: East and West. A depot will be set up in each zone to serve the respective districts and estates. The manual procedure will make routing decisions and vehicle assignment for estates within the same district. From the Operation Manager's experience, there were about three to at most six estates clustered in the same district. A messenger usually handles delivery to two or three housing estates in a day. The route sequence is determined by visiting estates within the same district with the largest bill volume first until the vehicle capacity or trip time limit is exceeded. This manual approach takes slightly less than an hour for the current problem set and the result is shown in Table 8. It is observed that the selected depot at $P3$ has exceeded its capacity limit by 849 bills.

The four algorithms are applied on the current instance and their performance with respect to time (iteration number) is shown in Fig. 4. The differences in performance are illustrated in this larger problem. SA arrives quickly at a good solution before the first half of the running time. Thereafter, the remaining time records no improvement. TA on its own has a comparatively slower improvement rate than SA. Applying one metaheuristic after another results in irregular improvement rate. However, small improvement is still possible after the switchover of method. In this instance, (TA, SA) is slightly superior among the four algorithms. However, the general performance of metaheuristics depends on the problem-specific instance, the computational time allowed, the tuning of parameters and the random nature of the probabilistic local search method. Table 9 shows the detailed location and routing results of (TA, SA). The loading procedure has not improved the current solution of one vehicle assigned to each

Table 7
Computational results on test problems

Problem	Solution value (total costs)					CPU time (s)				
	Branch and bound	(TA, SA)	(SA, TA)	TA	SA	Branch and bound	(TA, SA)	(SA, TA)	TA	SA
1	309,817	309,817	309,817	309,818	309,817	1155	0.55	0.49	0.44	0.49
2	309,808	309,808	309,808	309,808	309,808	982	0.55	0.66	0.54	0.49
3	312,036[a]	312,036	312,036	312,036	312,036	> 10,000	0.88	0.83	0.82	1.09
4	616,050	616,050	616,050	616,050	616,050	8,107	0.61	0.55	0.60	0.55
5	620,141[a]	620,165	620,165	620,165	620,165	> 10,000	0.88	0.88	1.21	0.77

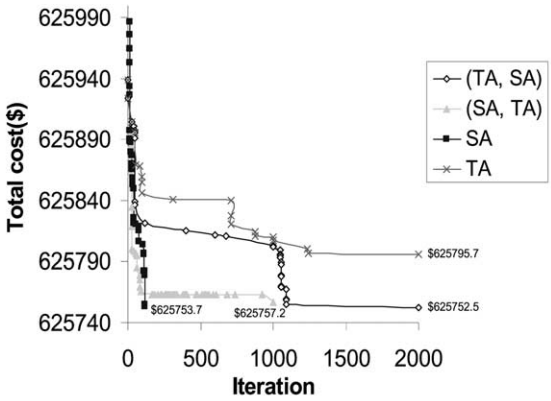
[a] Best solution found within the time limit of 10,000 CPU s.

Table 8
Depot location and routes assignment by manual approach

Depot	Zone	Route number	Estates	Route	Number of bills	Trip time (h)
*P*3	Kowloon East	1	*P*5, *P*6, *P*7	*P*3–*P*7–*P*6–*P*5–*P*3	3670	8.00
		2	*P*20, *P*21	*P*3–*P*20–*P*21–*P*3	3776	7.17
		3	*P*12, *P*13, *P*19	*P*3–*P*13–*P*3	4037	7.27
		4		*P*3–*P*19–*P*12–*P*3	3642	6.77
		5	*P*17, *P*18	*P*3–*P*17–*P*18–*P*3	3646	7.03
		6	*P*8, *P*24	*P*3–*P*8–*P*24–*P*3	3241	6.38
		7	*P*26, *P*27	*P*3–*P*27–*P*26–*P*3	1182	3.78
		8	*P*15, *P*16	*P*3–*P*15–*P*16–*P*3	2655	6.83
Total					25849[a]	53.23
*P*4	Kowloon West	9	*P*, 9*P*10, *P*11, *P*22, *P*23	*P*4–*P*9–*P*10–*P*11–*P*22–*P*23–*P*4	2036	7.50
		10	*P*28	*P*4–*P*28–*P*4	3301	7.62
		11	*P*29, *P*30, *P*31	*P*4–*P*30–*P*31–*P*4	3300	7.27
		12	*P*14, *P*25	*P*4–*P*25–*P*29–*P*14–*P*4	2567	7.22
Total					11204	29.61

[a] Depot capacity is exceeded by 849 bills.

established route. The cost comparison with the manual solution is given in Table 10. All algorithms are significantly better than the manual procedure and the pure descent procedure in terms of cost. The variable cost from the best solution provided by (TA, SA) achieves cost saving of 6% over the manual solution. In terms of productivity, one fewer messenger is required and one vehicle is saved. The computational time taken is about 6 CPU s on a Pentium III 866 as compared with slightly less than



Manual solution: $627,399 (produced in slightly less than an hour)
Descent procedure: $627,280 (same running time as the four metaheuristics)

Fig. 4. Performance of the four metaheuristics.

Table 9
Depot location and routes assignment by (TA, SA) heuristic

Depot	Route number	Estates	Route	Number of bills	Trip time (h)
*P*3	1	*P*6, *P*5, *P*9, *P*24	*P*3–*P*6–*P*5–*P*9–*P*24–*P*3	2370	7.15
	2	*P*19, *P*27, *P*26	*P*3–*P*19–*P*27–*P*26–*P*3	3619	7.52
	3	*P*, 21*P*20	*P*3–*P*21–*P*20–*P*3	3776	7.17
	4	*P*7	*P*3–*P*7–*P*3	2960	5.13
	5	*P*8, *P*12, *P*25	*P*3–*P*8–*P*12–*P*25–*P*3	3716	7.92
	6	*P*13	*P*3–*P*13–*P*3	4037	7.27
Total				20478	42.16
*P*4	7	*P*, 15*P*16, *P*23	*P*4–*P*15–*P*16–*P*23–*P*4	2799	7.15
	8	*P*17, *P*18, *P*14	*P*4–*P*17–*P*18–*P*14–*P*4	4016	7.97
	9	*P*29, *P*30	*P*4–*P*29–*P*30–*P*4	3519	7.92
	10	*P*11, *P*10, *P*31, *P*22	*P*4–*P*11–*P*10–*P*31–*P*22–*P*4	2940	7.95
	11	*P*28	*P*4–*P*28–*P*4	3301	7.62
Total				16575	38.61

an hour in the manual approach. The executable program (of size 465K bytes) can be simply run on Windows or DOS environment without other software requirement. This reduces future problems of software incompatibility.

Despite not having the optimal solution for exact performance evaluation, we have introduced some efficient metaheuristics for this practical, integrated logistics problem. The solution of which can be used to streamline the procedure of an exact branch and bound algorithm.

4.5. Management comments

The computerized algorithm has been tested by the operation team and is well perceived by the management. The choice of the depot locations at (*P*3,*P*4) is acceptable. Although the same set of

Table 10
Cost comparison between (TA, SA) heuristic and the manual solution

	Unit	Unit cost	Heuristic solution		Manual solution	
Cost item			Units required	Cost	Units required	Cost
Fixed cost						
Depot set up cost	Depot	$300,000	2	$600,000	2	$600,000
Variable cost						
Depot rental cost	Depot	$1800	2	$3600	2	$3600
Vehicle rental cost	Vehicle	$1200	11	$13,200	12	$14,400
Vehicle operating cost	Hour	$70	80.77	$5652.50	82.84	$5,799
Messenger salary	Messenger	$300	11	$3300	12	$3,600
Total variable cost				$25,752.50		$27,399

locations is found by the manual approach, the route assignments are different. Apart from cost savings and increased efficiency, it is observed that a more balanced load is achieved between depots, among messengers (or vehicles) and the average slack time of messenger is smaller. These results show that the algorithm achieves significant improvement over the manual method. The management further comments that the program is flexible for their decision-making, such as they can change the working hour parameter in the input when they consider employing part-time staff. The full data set for Hong Kong is now being collected to extend the scope of this study. This capacitated LRP with staff assignment (loading phase) is both a strategic and daily operational problem for the company. Similar to the practice in logistics industry, this telecommunication service company will consider incorporating information technology and geographic information systems to monitor the daily performance and provide a more personal delivery service to individual customers in future.

5. Conclusion

We consider a practical integrated problem with location, routing and loading decisions. A simultaneous approach in handling these interdependent decisions is adopted. Capacity restrictions on vehicles, facilities and trip time limit exist. Most LRP literature does not consider all such complexities. Exact approach in past literature is usually proposed for some simpler unconstrained problems. Practical problems are likely more complex and large in size. The metaheuristic models proposed here have been tested on some smaller problems to give fast and near optimal solutions. In particular, the hybrids shed new light for further investigation. On the other hand, the general branch and bound framework developed for this integrated problem can be further improved by incorporating some special procedures in the exact approach for simpler problems. Exact solutions will be useful for benchmarking purpose.

The metaheuristic approach is well perceived by management and is shown to generate cost savings for the company. We are currently extending the scope of the study to consider the full problem size of Hong Kong. The integration of optimization models with information and communication systems is likely to be important for the future logistics development of the society.

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